

UPI FAVOURITISM

DATA ANALYSIS ON USER PREFERENCE

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UPI and Data Sample Overview

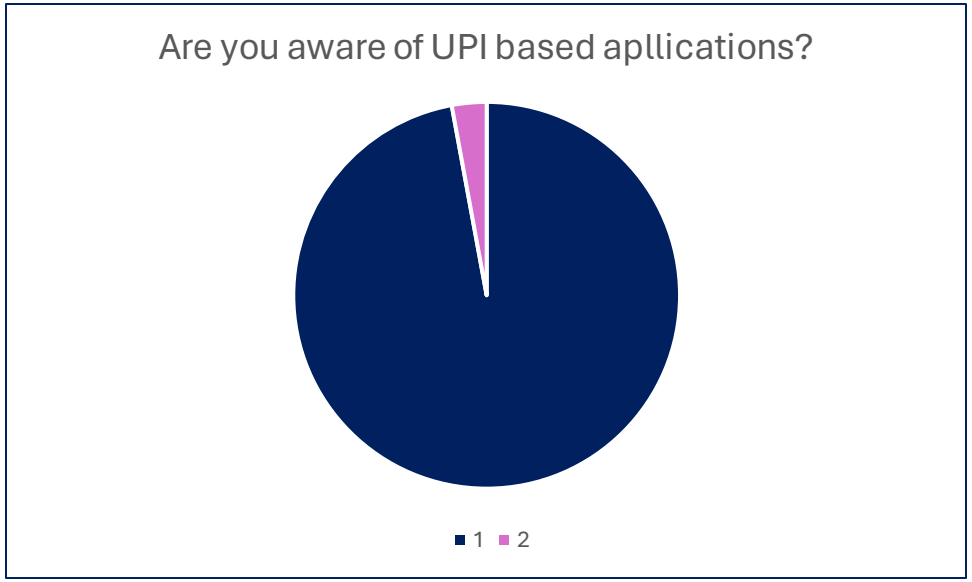
UNIFIED PAYMENTS INTERFACE

- UPI: Unified Payments Interface by NPCI.
- Enables instant bank transfers via mobile.
- Uses VPAs or QR codes for transactions.
- Supported by major Indian banks.
- Revolutionizes digital payments, enhancing convenience and security.

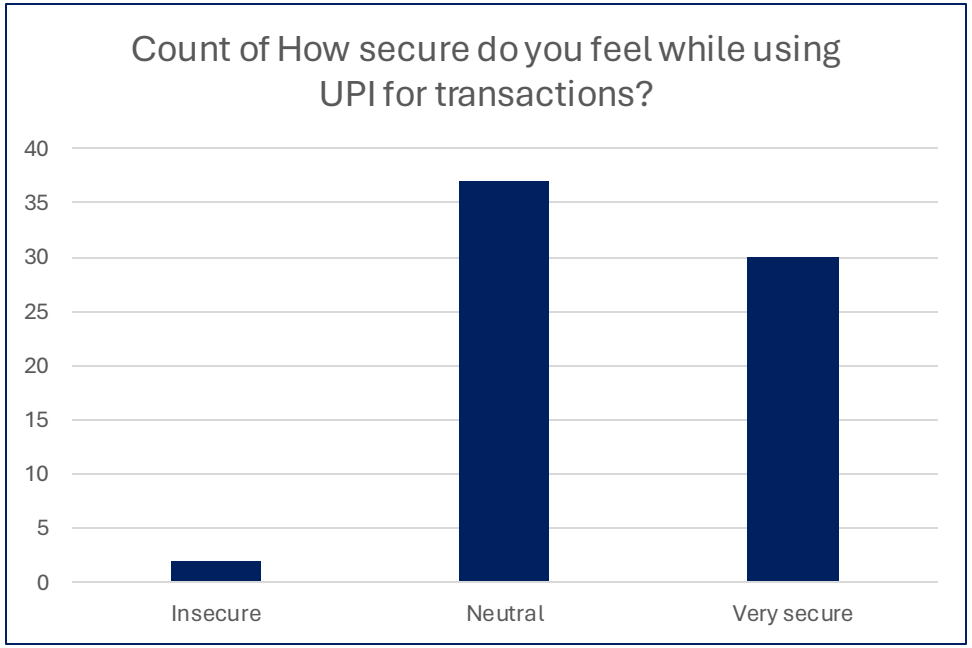


- Our study comprises a sample size of **69 individuals**, carefully selected to represent diverse age groups.
- The participants are categorized into three age brackets**: 17-20, 20-25, and 25 and above**.
- Analysis of the responses indicates that 81.2% of the participants fall within the 17-20 age group, highlighting a predominant representation in this demographic.
- Additionally, 11.6% of the participants belong to the 20-25 age range, while 7.2% are aged 25 and above.
- Notably, the majority of our sample consists of students from renowned educational institutions such as Lady Shri Ram College (LSR) and other colleges affiliated with Delhi University (DU).

Basic Graphs and their analysis



In the provided graph, a value of 1 represents "yes," indicating that 95.71% of the total sample population is aware of UPI-based applications. Conversely, a value of 2 signifies "no," indicating that only 2.86% of the sample is unaware of such applications.

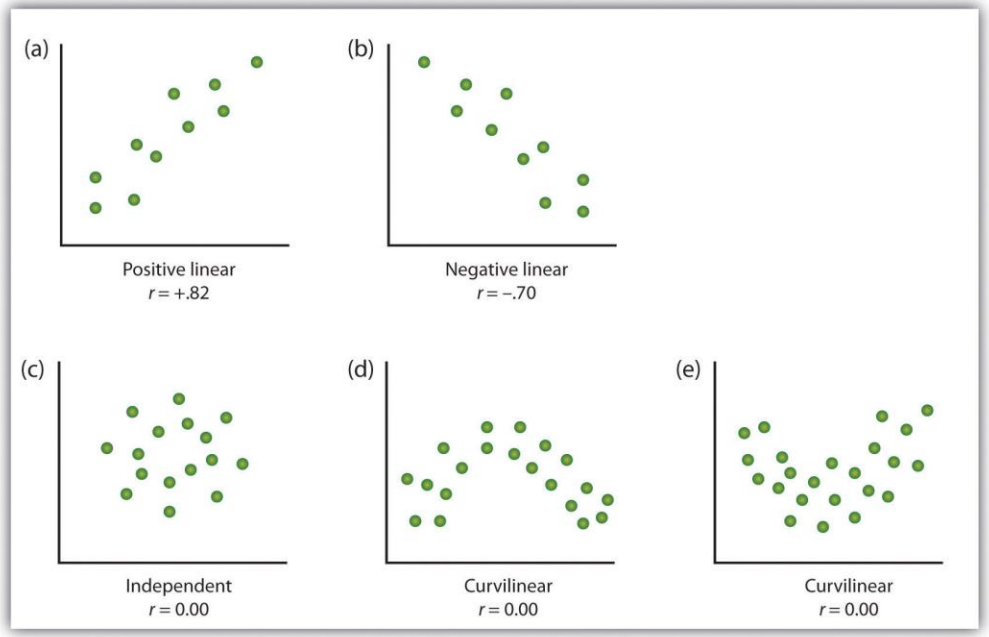


Among respondents, 50% expressed a neutral stance regarding their perceived security when using a UPI app. Additionally, 43.47% reported feeling highly secure, while a minority, constituting 2.89%, expressed feelings of insecurity regarding UPI app usage.

CORRELATION FORMULAS USED

PARTIAL CORRELATION

- **Partial correlation** measures the strength and direction of the relationship between two variables while controlling for the influence of one or more additional variables.
- For example, we could calculate the partial correlation between age and satisfaction with **UPI-based applications** while controlling for factors like frequency of usage, perceived security, and usage purposes.
- This analysis helps to understand the unique contribution of each variable to the correlation between age and satisfaction, accounting for the effects of other variables.



$$r_{13.2} = \frac{[r_{13} - (r_{12})(r_{23})]}{\sqrt{1-(r_{12})^2}\sqrt{1-(r_{23})^2}}$$

MULTIPLE CORRELATION

- **Multiple correlation** examines the relationship between one variable (dependent variable) and two or more other variables (independent variables) simultaneously.
- In the context of **UPI-based applications**, we could examine the multiple correlation between overall satisfaction with UPI and a combination of independent variables such as age, frequency of usage, perceived security, reliability ratings, usage purposes, and satisfaction with customer support.
- This analysis helps to understand how well a combination of independent variables predicts overall satisfaction with UPI-based applications.

$$R_{1.23} = \frac{\sqrt{r^2_{12} + r^2_{13} - 2\, r_{12}\, r_{13}\, r_{23}}}{\sqrt{1 - r^2_{23}}}$$

$$R_{2.13} = \frac{\sqrt{r^2_{21} + r^2_{23} - 2\, r_{21}\, r_{23}\, r_{13}}}{\sqrt{1 - r^2_{13}}}$$

$$R_{3.12} = \frac{\sqrt{r^2_{31} + r^2_{32} - 2\, r_{31}\, r_{32}\, r_{12}}}{\sqrt{1 - r^2_{12}}}$$

CORRELATION ANALYSIS

Age Distribution: Most respondents are in the age group of **17-20,** with a few in the age groups of 20-25 and 25

Usage Purposes: Peer-to-peer transfers, online shopping, bill payments, and in-store QR code scans are the most common uses of UPI apps.
Some respondents use UPI for specific purposes like **zomato & swiggy** orders.

Transaction Limit Perception: Respondents' perception of the maximum transaction limit for UPI payments varies, with some stating **50,000** and others mentioning **1 lakh.**

Awareness and Usage of UPI:
Almost all respondents are aware of UPI-based applications.
Daily usage of UPI apps is more common, with some using them weekly or rarely.
Google Pay and PhonePe are popular among the respondents.

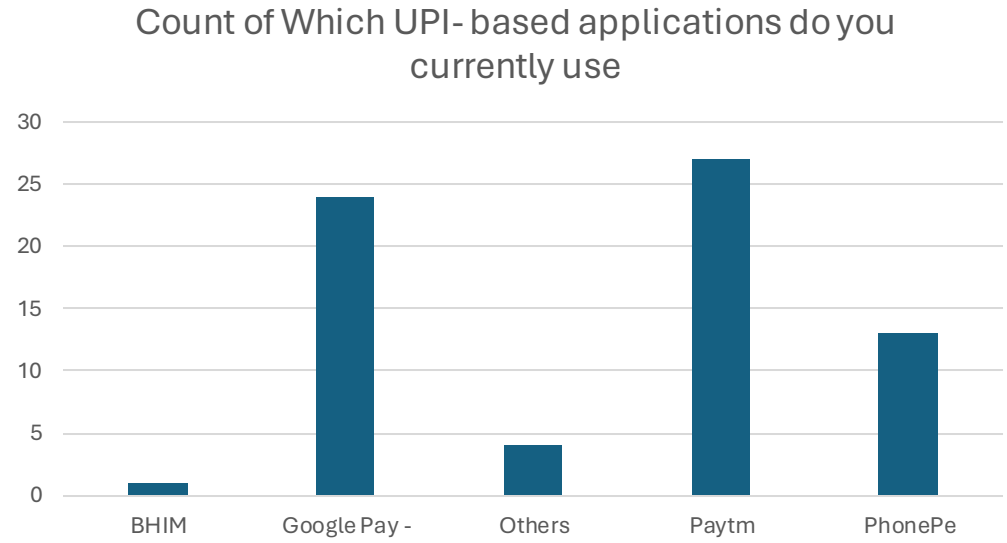
Factors Influencing Choice: Security, ease of use, user interface, and customer service are key factors influencing the choice of using a **UPI-based application.**

Perceived Security and Reliability:
Perceptions of security and reliability vary among respondents, with some feeling very secure while others are neutral or insecure.
Ratings for overall reliability range from **0 to 10**, indicating varying levels of confidence in UPI transactions.

Reasons for Non-Usage:
Technical difficulties and fraud concerns are cited as prevalent reasons for some people not using UPI.

Dream UPI App Features: Respondents desire features like security, speed, user-friendliness, customer support, credit system, and **EMI options i**n their dream UPI app.

Satisfaction with Customer Support:
Satisfaction levels with customer support provided by UPI-based applications vary among respondents.



How secure do you feel while using UPI for transactions?

How frequently do you use UPI-based applications?	1
How secure do you feel while using UPI for transactions?	0.098614534

MULTIPLE CORRELATION ANALYSIS

upi reliability rating(x1)	rate your satisfaction(x2)	payment limit on upi transactions(x3)

sd(x1)	1.532818031
sd(x2)	1.058598536
sd(x3)	25069.36547

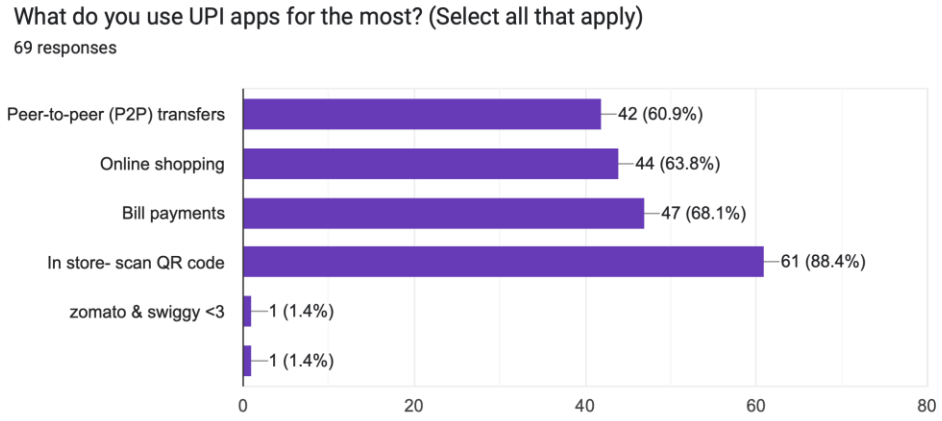
cov(x1,x2)	0.7609745852
cov(x2,x3)	6714.975845
cov(x1,x3)	7222.222222

R1.23	**0.5289798205**
R2.13	**0.5493137884**
R3.12	**0.3604705844**

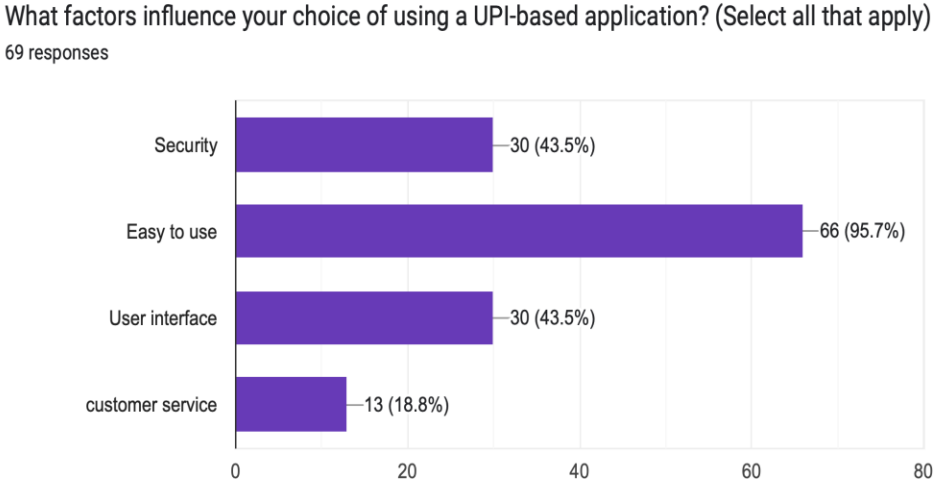
corr(x1,x2)	0.4758701251
corr(x2,x3)	0.2558617148
corr(x1,x3)	0.1900519751

	r12.3	r13.2
	0.4272431008	0.06829502887
1-r13^2	0.9638802468	0.773547624
1-r23^2	0.9345347829	
	0.9007796172	
^1/2	0.9490941034	0.8502394727
final	**0.4501588402**	**0.08032446277**

Basic Graphs and their analysis

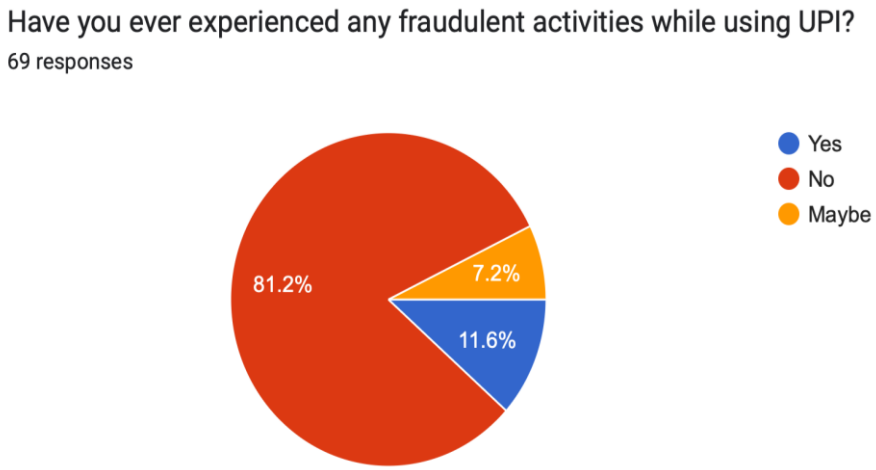


Based on the provided data, it can be inferred that 88.4% of people utilize UPS for in-store QR code scanning, while the least usage is observed for Sonata and Swiggy, with only 1.4% of individuals using these services. This suggests a significant preference for UPS in facilitating in-store QR code interactions, contrasting sharply with the minimal engagement with Sonata and Swiggy.

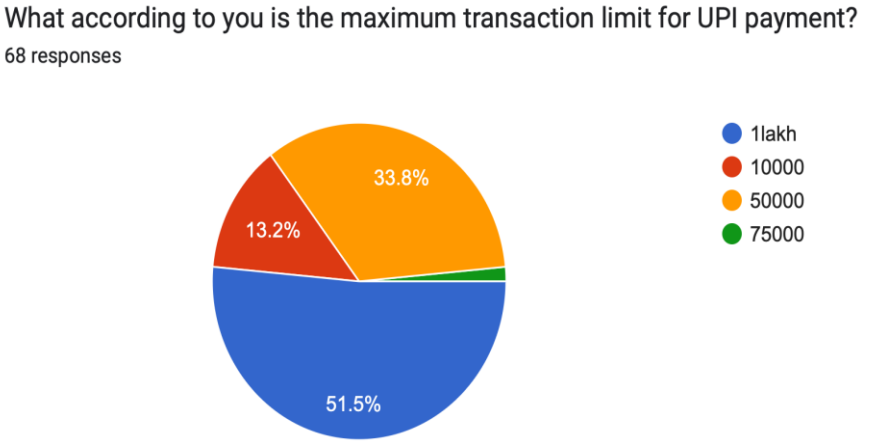


Based on the aforementioned data, it can be inferred that 95.7% of people utilize UPI applications due to their ease of use, followed by security and user interface at 43.5%. This indicates a strong preference for UPI apps primarily driven by their user-friendly nature, with security and interface quality also playing significant roles albeit to a lesser extent.

Basic Graphs and their analysis



Based on the provided data, it is evident that 81.2% of people have never experienced any fraudulent activities while using UPI, while 11.6% have encountered such incidents. This suggests that a majority of users have had a smooth experience with UPI transactions, with a notable minority having faced fraudulent activities.



From the data provided, it is apparent that 50% of the sample population answered the question correctly, stating 1 lakh as the maximum limit for UPI transactions, while the other half answered incorrectly. This highlights the necessity for more financial education, as it indicates a lack of awareness among a significant portion of the surveyed individuals regarding the transaction limits associated with UPI.

UPI Favouritism

Hey,
We are first years statistics students, this form will help us in our project research, we have tried to keep if short and to the point. Hope you can take 60 secs to fill it.
Thanks✨

1. Age

   Mark only one oval.

   - ◯ 17-20
   - ◯ 20-25
   - ◯ 25+

2. Are you aware of UPI-based applications?

   Mark only one oval.

   - ◯ Yes
   - ◯ No
   - ◯ Maybe

3. Which UPI-based application(s) do you currently use? (Select all that apply)

   Mark only one oval.

   - ◯ Google Pay -
   - ◯ PhonePe
   - ◯ Paytm
   - ◯ BHIM
   - ◯ Others

4. How frequently do you use UPI-based applications?**

Mark only one oval.

- daily
- Weekly
- Monthly
- Rarely
- Never

5. How secure do you feel while using UPI for transactions?

Mark only one oval.

- Very secure
- Insecure
- Neutral

6. On a scale of 1-10 (with 10 being the highest), how would you rate the overall reliability of U in processing transactions?

Mark only one oval.

1	2	3	4	5	6	7	8	9	10
◯	◯	◯	◯	◯	◯	◯	◯	◯	◯

7. What do you use UPI apps for the most? (Select all that apply)

Tick all that apply.

- [ ] Peer-to-peer (P2P) transfers
- [ ] Online shopping
- [ ] Bill payments
- [ ] In store- scan QR code
- [ ] Other: _____

8. What do you think is/are the most prevalent reason(s) for some people not using UPI? (Sel all that apply

Tick all that apply.

- [ ] Finding it difficult due to technicalities
- [ ] Not having internet availability all the time
- [ ] Fraud concerns
- [ ] Other: _____

9. What factors influence your choice of using a UPI-based application? (Select all that apply)

Tick all that apply.

- [ ] Security
- [ ] Easy to use
- [ ] User interface
- [ ] customer service
- [ ] Other: _____

10. What according to you is the maximum transaction limit for UPI payment?

Mark only one oval.

- 1lakh
- 10000
- 50000
- 75000

11. Have you ever experienced any fraudulent activities while using UPI?

Mark only one oval.

- Yes
- No
- Maybe

12. Imagine creating your dream UPI app. What are the key ingredients (security, speed, user friendliness, etc.) you'd need to make it the perfect recipe for managing your money?

13. How satisfied are you with the customer support provided by UPI-based applications?

Mark only one oval.

	1	2	3	4	5	
Diss	◯	◯	◯	◯	◯	Satisfied

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